A function is an equivalence exactly when it is both surjective and an embedding

The Univalent Foundations Program

hott-book:equivalences.tex:local-snapshot-20261011; CC BY-SA 3.0 Unported

Editorial note. The counted statement is the author's 'thm:mono-surj-equiv' (source lines 571-573, Theorem 4.6.3); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. The retained counter values (chapter 4, section 6, theorem offset 2) make the counted theorem print as Theorem 4.6.3, matching the original book numbering taken from the label-number dump of the pinned snapshot.

Editorial note. No context span is retained: the statement and proof are self-contained apart from one labelled invocation.

Editorial note. This source directory intentionally contains only the author file, so the retained book macro library is bundled as a per-task copy inside this task directory and referenced from preamble_spans; the copy is verbatim from the same pinned book snapshot and no macro is added, changed or invented.

Editorial scope. One original result of the HoTT Book, counted once: equivalences are exactly the surjective embeddings (Theorem 4.6.3).

Allowed standard prerequisites (editorial).

- Surjective map. f is surjective when trunc(-1)(fiber_f b) holds for every b.. It is one of the two hypotheses of the converse direction and the conclusion of the forward direction.
- Embedding. f is an embedding when ap f : (x = y) -> (f x = f y) is an equivalence for all x, y.. It is used to produce r : x = y with ap f (r) = p 'ct' opp q.
- Paths in a sigma type. (x,p) = (y,q) in a sigma type is equivalent to a pair (r, trans r p = q).. It rearranges the equality at source line 584 into the transport statement.

Source reference. Original theorem 2.11.1: sources/hott-hits/book-labelnumbers-first-edition.tex, source lines 1854–1854. Original result cited by the retained text but located outside the retained span. It is a navigation pointer only: the counted statement's own author proof is retained line for line and no part of it is replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

Theorem 4.6.3. *A function $f : A \to B$ is an equivalence if and only if it is both surjective and an embedding.*

Proof. If f is an equivalence, then each $\mathsf{fib}_f(b)$ is contractible, hence so is $\|\mathsf{fib}_f(b)\|$, so f is surjective. And we showed in original theorem 2.11.1 that any equivalence is an embedding.

Conversely, suppose f is a surjective embedding. Let $b : B$; we show that $\sum_{(x:A)}(f(x) = b)$ is contractible. Since f is surjective, there merely exists an $a : A$ such that $f(a) = b$. Thus, the fiber of f over b is inhabited; it remains to show it is a mere proposition. For this, suppose given $x, y : A$ with $p : f(x) = b$ and $q : f(y) = b$. Then since ap_f is an equivalence, there exists $r : x = y$ with $\mathsf{ap}_f(r) = p \cdot q^{-1}$. However, using the characterization of paths in Σ-types, the latter equality rearranges to $r_*(p) = q$. Thus, together with r it exhibits $(x, p) = (y, q)$ in the fiber of f over b. □